

Ligjerata 8dh

Teorema e Toricelit

$$\frac{\rho v_1^2}{2} + \rho g h_1 + p_1 = \frac{\rho v_2^2}{2} + \rho g h_2 + p_2$$

$$h_1 = 0$$

$$h_2 = h$$

$$p_1 + \frac{\rho v_1^2}{2} = p_2 + \frac{\rho v_2^2}{2} + \rho g h$$

$$p_1 = p_2 = p_a$$

$$\frac{\rho v_1^2}{2} - \frac{\rho v_2^2}{2} = \rho g h$$

$$\cancel{\rho} \frac{(v_1^2 - v_2^2)}{2} = \cancel{\rho} g h$$

$$v_1^2 - v_2^2 = 2 g h$$

$$v_1^2 = v_2^2 + 2 g h \quad v_1 \gg v_2$$

$$v_1^2 = 2 g h \quad | \sqrt{}$$

$$v_1 = \sqrt{2 g h}$$

